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Markov chain and MCMC Examples

CSC1111
Building Complex Computational Models

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Markov Process

Example - Soft Drink Market Share :

- In a soft drinks market there are two Brands: Coke & Pepsi.
- At any time n sample space of possible outcomes is (P, C)
- Here P = % market share that is Pepsi's in one year and C , = % that is Coke's.
- Know that chance of switching from Coke to Pepsi is 0.1
- And the chances of someone switching from Pepsi to Coke are 0.3.
- How can the changes in the different proportions be modelled?

Assume that Coke is the soft drink to start with.

MCMC examples with code

- A simple Metropolis-Hastings MCMC with R code:

<https://theoreticalecology.wordpress.com/2010/09/17/metropolis-hastings-mcmc-in-r/>

- My first MCMC sampler:

http://sbfnk.github.io/mfiidd/mcmc.html#4_my_first_mcmc_sampler

- Interactive MCMC examples where you can change parameters values for different samplers

<https://github.com/chi-feng/mcmc-demo>

- R examples with solutions and plots:

https://www.casact.org/sites/default/files/2021-03/masii_complete_textref.pdf

More MCMC examples with code

- Some interesting online resources and examples:

1. <https://twiecki.io/blog/2014/01/02/visualizing-mcmc/>
2. https://betanalpha.github.io/assets/case_studies/markov_chain_monte_carlo.html#6_putting_it_all_into_practice
3. <https://towardsdatascience.com/markov-chain-monte-carlo-in-python-44f7e609be98>